Step 3 (Assignment) :

		Men			
		1	2	3	d
Task	*P*	[0]	6	9	~~[0]~~
	Q	4	7	[0]	~~[0]~~
	R	26	[0]	9	~~[0]~~
	S	9	10	14	[0]

Since each row and each columncontains exactly one assignment,all the three men have been assigned a task. But task *S* is not assigned to any Man. The optimal assignment schedule and total cost is

Task	Men	cost
P	1	9
Q	3	6
R	2	20
S	*d*	0
Total cost		35

The optimal assignment (minimum) cost = ₹ 35.

Exercise 10.2

1. What is the Assignment problem?
2. Give mathematical form of Assignment problem.
3. What is the difference between Assignment Problem and Transportation Problem?
4. Three jobs A, B and C one to be assigned to three machines U, V and W. The processing cost for each job machine combination is shown in the matrix given below. Determine the allocation that minimizes the overall processing cost.

		Machine		
		U	*V*	*W*
Job	*A*	17	25	31
	B	10	25	16
	C	12	14	11

(cost is in ₹ per unit)

5. A computer centre has got three expert programmers. The centre needs three application programmes to be developed. The head of the computer centre, after studying carefully the programmes to be developed, estimates the computer time in minitues required by the experts to the application programme as follows.

		Programmes		
		P	*Q*	*R*
Programmers	1	120	100	80
	2	80	90	110
	3	110	140	120

Assign the programmers to the programme in such a way that the total computer time is least.

6. A departmental head has four subordinates and four tasks to be performed. The subordinates differ in efficiency and the tasks differ in their intrinsic difficulty. His estimates of the time each man would take to perform each task is given below:

		Tasks			
		1	2	3	4
Sub-ordinates	*P*	8	26	17	11
	Q	13	28	4	26
	R	38	19	18	15
	S	9	26	24	10

How should the tasks be allocated to subordinates so as to minimize the total man-hours?

7. Find the optimal solution for the assignment problem with the following cost matrix.

		Area			
		1	2	3	4
Salesman	*P*	11	17	8	16
	Q	9	7	12	6
	R	13	16	15	12
	S	14	10	12	11

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